

Voice Synthesis, Transformations, and Sound Effects: A portfolio

Cumhur Erkut,

with help of Christie Laurent, Juan Alonso, David Südholt, and Anders R Børgum

Intro

- Started physically on aging formants
- Benefited in DL disentanglements in DDSP, RAVE -> SRAVE, back to music

Methods

- Tunable Voice Aging Model
- MFCC playback using Z in DDSP
- DDSP cross-synthesis, plugins
- Full embeddings in S-RAVE

Results



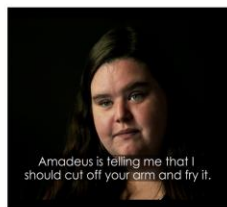
- Significant performance: RT from 1 to 15, transformation data from 20 minutes to 30 seconds
- Concerns: Voice spoofing

Discussion

- Diverse portfolio (Hearing Aids, XR, Robotics, Embedded AI)
- PINNS <-> Transformers
- Active Inference (Bayesian)
- Evaluation: Audio Aesthetics?

Interplay between sound, music, and speech. Transformers for embeddings, diffusion for real-time synthesis.

PROBLEM



- 30% of patients with schizophrenia treatment-resistant Meltzer HV. (1997)
- those whose symptoms get better from medication experience terrible side-effects and often discontinue taking their meds Lieberman, J. A., et al. (2005)
- This is the most expensive disease to treat in psychiatry Kadakia A. et al. 2022, Christensen, M.K. et al. 2022, FosseehA. et al. 2018,

Amadeus is telling me that I should cut off your arm and fry it.

AVATAR THERAPY

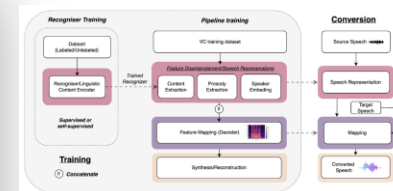


The patient together with the therapist create the avatar of the malevolent voice



Scan to hear more at cerkut.github.io, click on photos to read more about talented Christie Laurent, Juan Alonso, David Südholt, and Anders R Børgum

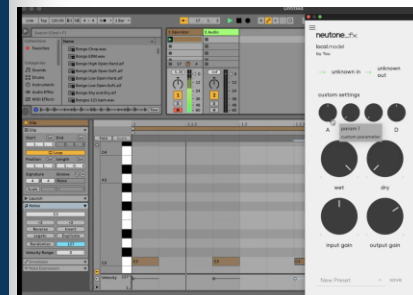
Recent Work



General Voice Conversion pipeline: S-RAVE



Commercial deployment possible: BabyAudio Humanoid



Back to musical instruments

